Universität Bonn

Mathematisches Institut
Prof. Dr. Shuntaro Yamagishi
Number Theory I
Summer 2026

Lecture Notes as of 24th April 2026

LaTeX by Hannes Heimberger

Corrections and Remarks

If you find a mistake in this file or have a remark concerning this document, I would be glad if you let me know. It would be great if you wrote a mail. I can be reached through the address `hannes.heimberger@uni-bonn.de`.

Solutions to exercises

I will add solutions to the exercises after their correction, at least if my tutor marked them correct. These may, however, not be written down in the most optimal way. The proofs left as exercsises will be marked as such. If you have suggestions about how to improve these proofs, comments are welcome.

Colours

My annotations to the notes are blue. Additional details that are neither part of the lecture nor relevant to the exam will be coloured in orange.

Thanks

I thank Daniel Emse for providing his template, which I adapted for these notes.

Contents

Number Theory I

1 Geometry of Numbers **2**
1.1 Introduction 2
1.2 Lattices and discrete subgroups 4
1.3 Convex Bodies 7
1.4 Minkowski's First Theorem 8
1.5 Minkowski's Second Theorem 13

Appendix

List of Definitions **A**

List of Theorems **B**

List of Lemmata **C**

Index **D**

Number Theory I

Notations and conventions

Number sets

- The natural numbers, denoted by $\mathbb{N}$, contain 0 (which is in accord with *DIN 5473* (*German Industrial Norm*) – Welcome to Germany! xD).

Subsets and -structures

- If we write $A \subset B$, we mean $A \subseteq B$ and $A \neq B$ (see *DIN 5473*).
- We write $\sqsubset$ or $\sqsubseteq$ for substructures (subfields, linear subspaces of vector spaces, etc.) of algebraic structures (didn't yet make it into *DIN 5473* :().

For $f : \mathbb{C} \times \cdots \to \mathbb{C}$, we define

$$N(f) = \{s \in \mathbb{C} : f(s, -) = 0\}$$

and

$$N^*(f) = \{s \in [0, 1] \times \mathbb{R} \subset \mathbb{C} : f(s, -) = 0\}.$$

Quantors

- If we write $\mathbb{V}_{\ldots}$, we mean something holds for almost all ...

Functions

- We write $(x, y) = \gcd(x, y)$.
- We have $\{x\} = x - \lfloor x \rfloor \ \ \mathbb{V}_{x \in \mathbb{R}}$.

Fields

- We write $\mathbb{A}$ for $\overline{\mathbb{Q}}$.

Chapter 1

Geometry of Numbers

1.1 Introduction

Definition 1.1.1 – $\mathbb{Z}^2_{\text{prim}}$

We denote

$$\mathbb{Z}^2_{\text{prim}} = \{(x, y) \in \mathbb{Z}^2 : (x, y) = 1\}.$$

Theorem 1.1.2 (Dirichlet, 1842)

Let $\alpha \in \mathbb{R}$. Then, for every integer $Q \geq 2$, there exists $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $0 < y \leq Q$ and $|x - \alpha y| \leq 1/Q$.

Proof. Consider the $Q+1$ numbers $0, \alpha - \lfloor \alpha \rfloor, 2\alpha - \lfloor 2\alpha \rfloor, \ldots, Q\alpha - \lfloor Q\alpha \rfloor$. Each of these is in the interval $[0, 1)$. Partition the interval $[0, 1)$ into Q subintervals of length $1/Q$. Now, we can apply the pigeonhole principle to deduce that

$$\exists_{0 \leq i \leq j \leq Q} : |(j\alpha - \lfloor j\alpha \rfloor) - (i\alpha - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

This implies

$$|(j - i)\alpha - (\lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor)| \leq \frac{1}{Q}.$$

Let $\tilde{y} = j - i$ and $\tilde{x} = \lfloor j\alpha \rfloor - \lfloor i\alpha \rfloor$. Then $|\tilde{y}\alpha - \tilde{x}| \leq 1/Q$. Let $d = (\tilde{y}, \tilde{x})$, $y = \tilde{y}/d$ and $x = \tilde{x}/d$ and divide through by d to get that

$$|y\alpha - x| \leq \frac{1}{dQ} \leq \frac{1}{Q}.$$

□

A very simple consequence of this:

Corollary 1.1.3 (Irrational approximation)

Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$. Then there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\left| \alpha - \frac{x}{y} \right| \leq \frac{1}{y^2}.$$

Proof. For each $Q \geq 2$, let $(x_Q, y_Q) \in \mathbb{Z}^2_{\text{prim}}$ be such that $0 < y_Q \leq Q$ and $|x_Q - y_Q\alpha| \leq {}^1/_Q$, which we know to exist from Theorem 1.1.2. This implies

$$\left|\alpha - \frac{x_Q}{y_Q}\right| \leq \frac{1}{Qy_Q} \leq \frac{1}{y_Q^2}.$$

Suppose the set

$$S = \{(x_Q, y_Q) : Q \geq 2\}$$

is finite. Then there exists $(x_0, y_0) \in two$ such that $(x_Q, y_Q) = (x_0, y_0) \quad \forall_{Q \geq Q_0}$ for some Q_0. This means

$$\left|\alpha - \frac{x_Q}{y_Q}\right| = \left|\alpha - \frac{x_0}{y_0}\right| \leq \frac{1}{Q} \quad \forall_{Q \geq Q_0}.$$

Taking the limit $Q \to \infty$ gives

$$\alpha = \frac{x_0}{y_0},$$

which is a contradiction due to the fact that α was supposed to be irrational. ↯ □

Question 1.1.4 (Is this error term optimal?)

Can we do better than ${}^1/_{y^2}$ for some $\alpha \in \mathbb{R}^2$? I.e. do there exist infinitely many $(x, y) \in \mathbb{Z}^2_{\text{prim}}$ such that $y > 0$ and

$$\exists_{\delta > 0} : \left|\alpha - \frac{x}{y}\right| < \frac{1}{y^{2+\delta}}.$$

Answer. Yes, but it's rare. However, this is not the case if α is algebraic over $\mathbb{Q}$. □

Theorem 1.1.5 (Roth, 1955)

Let $\alpha \in \mathbb{A}$ and $\delta > 0$. Then

$$\#\{(x, y) \in \mathbb{Z}^2_{\text{prim}} : y > 0, |\alpha - {}^x/_y| \leq {}^1/_{y^{2+\delta}}\} < \aleph_0.$$

Proving this is one of the main goals of this course. Another important result we will prove is Thue's theorem.

Theorem 1.1.6 (Thue, 1909)

Let $F \in \mathbb{Z}[X, Y]$ be a binary form (hence homogenious) of degree d. Suppose $\deg F(X, 1) = d$ and that $F(X, 1)$ has at least three distinct zeroes in $\mathbb{C}$. Let $m \in \mathbb{Z} \setminus \{0\}$. Then

$$\#\{(x, y) \in \mathbb{Z}^2 : F(x, y) = m\} < \aleph_0.$$

Also, time permitted, Professor Yamagishi'd like to go through the details of Stanley

Yao Xiao's proof of Büchi's problem.[1]

Problem 1.1.7 (Büchi)

Let $0 < x_1 < \cdots < x_5$ be integers such that

$$x_{i+1}^2 - 2x_i^2 + x_{i-1}^2 = 2 \ \forall_{i \in \{2,\ldots,4\}}.$$

Then there exists $x_0 \in \mathbb{Z}$ such that $x_i = x_0 + i \ \forall_{1 \leq i \leq 5}$.

This leads to the following refinement of Hilbert's tenth problem: There is no algorithm which decides the existence of integer solutions to an arbitrary system of diagonal quadratic equations:

$$A \begin{pmatrix} x_1^2 \\ \vdots \\ x_n^2 \end{pmatrix} = b.$$

Professor Yamagishi recommends the book **An introduction to the geometry of numbers** by Cassels. He also recommend a course on **Prime Numbers** taught next semester (Selected Topics in Number Theory). Tutorial times are **Monday at 14:00** and **Tuesday at 12:00**. You can register for these **Tomorrow at 16:00**.

Lecture 1 (15th April 2026)

. . . . . .

Lecture 2 (17th April 2026)

1.2 Lattices and discrete subgroups

Let $n \geq 2$. Given $x_1, \ldots, x_n \in \mathbb{R}^n$, we say that they are linearly independent over $\mathbb{R}$, if $c_1x_1 + \cdots + c_rx_r \neq 0 \ \forall_{(c_1,\ldots,c_r)^t \in \mathbb{R}^r \setminus \{0\}}$. We consider the usual topology on $\mathbb{R}^n$ generated by the Euclidean metric. For subsets of $\mathbb{R}^n$, we consider the subspace topology induced by this: We say a subset $\Lambda \subseteq \mathbb{R}^n$ is **discrete** if $\#\Lambda \cap B < \aleph_0$ for every bounded subset $B \subseteq \mathbb{R}^n$. Furthermore, Λ is a **discrete subgroup** if it is discrete and $c_1x_1 + c_2x_2 \in \Lambda \ \forall_{c_1,c_2 \in \mathbb{Z}, x_1,x_2 \in \Lambda}$. In this case, we define $\mathrm{rank}(\Lambda) = r$ as the maximal number r such that Λ contains r linearly independent vectors over $\mathbb{R}$. A **basis** of Λ is a set $\{v_1, \ldots, v_r\}$ of linearly independent vectors over $\mathbb{R}$ such that $\Lambda = \{c_1v_1 + \cdots + c_rv_r : c_1, \ldots, c_r \in \mathbb{Z}\}$.

Lemma 1.2.1 (Lattice Basis Construction)

Let Λ be a discrete subgroup of $\mathbb{R}^n$ and $\mathrm{rank}(\Lambda) = r \geq 1$. Also assume that $\{w_1, \ldots, w_r\} \subseteq \Lambda$ is linearly independent over $\mathbb{R}$. Then there exists a basis $\{v_1, \ldots, v_r\}$ of Λ such that for all $1 \leq k \leq r$, we have

$$v_k = \alpha_{k,1}w_1 + \cdots + \alpha_{k,k}w_k \ \forall_{\alpha_{k,1},\ldots,\alpha_{k,k} \in [0,1] \text{ and } \alpha_{k,k} \neq 0}.$$

Proof. Let $1 \leq k \leq r$. We define

$$S_k = \{\alpha_1w_1 + \cdots + \alpha_kw_k : \alpha_1, \ldots, \alpha_k \in [0,1], \alpha_k \neq 0\}.$$

[1] Professor Yamagishi: This is a paper by a friend of mine. Actually, it has nothing to do with geometry of numbers. But I want to do it anyway.

Since Λ is discrete and S_k is bounded,

$$\underbrace{0 <}_{w_k \in \Lambda \cap S_k} \#(\Lambda \cap S_k) \ll 1.$$

We choose v_k to be a vector from $\Lambda \cap S_k$ with the minimal α_k. Let us write

$$v_k = \alpha_{k,1} w_1 + \cdots + \alpha_{k,k} w_k.$$

Then for any

$$\alpha_1 w_1 + \cdots + \alpha_k w_k \in \Lambda \cap S_k,$$

we have $0 < \alpha_{k,k} \leq \alpha_k \leq 1$. It is clear that $\{v_1, \ldots, v_r\}$ is linearly independent over $\mathbb{R}$. For each $0 \leq k \leq r$, let

$$M_k = \Lambda \cap \left\{ \sum_{j=1}^{k} \alpha_j w_j : \alpha_1, \ldots, \alpha_k \in \mathbb{R} \right\} \text{ with } M_0 = \{0\}.$$

Note that $M_r = \Lambda$. We now use induction to show that $M_k = \{c_1 v_1 + \cdots + c_k v_k : c_1, \ldots, c_k \in \mathbb{Z}\}$ $\forall_{0 \leq k \leq r}$. When $k = 0$, the statement is trivially true. Suppose the statement holds for $k - 1$.
Let

$$x = \sum_{j=1}^{k} \beta_j w_j \in M_k \subseteq \Lambda \text{ and set } z_k = \left\lfloor \frac{\beta_k}{\alpha_{k,k}} \right\rfloor \in \mathbb{Z}.$$

Then, $0 \leq \beta_k - z_k \alpha_{k,k} < \alpha_{k,k}$. Consider $x - z_k \underbrace{v_k}_{\in \Lambda} \in \Lambda$. Also,

$$\begin{aligned} x - z_k v_k &= \sum_{j=1}^{k} (\beta_j - z_k \alpha_{k,j}) w_j \\ &= \underbrace{\sum_{j=1}^{k} \lfloor \beta_j - z_k \alpha_{k,j} \rfloor w_j}_{\in \Lambda} + u, \end{aligned}$$

where

$$u = \sum_{j=1}^{k-1} \{\beta_j - z_k \alpha_{k,j}\} w_j + (\beta_k - z_k \alpha_{k,k} w_k).$$

It is clear that $u \in \Lambda$. Also, if $\beta_k - z_k \alpha_{k,k} > 0$, then $u \in S_k$ (because $\{\beta_j - z_k \alpha_{k,j}\} \in [0, 1]$ and $0 < \beta_k - z_k \alpha_{k,k} < \alpha_{k,k} \leq 1$), which implies $u \in \Lambda \cap S_k$. But this contradicts the minimality of $\alpha_{k,k}$ $\lightning$. Hence, $\beta_k = z_k \alpha_{k,k}$. Then,

$$x - z_k v_k = \sum_{j=1}^{k-1} (\beta_j - z_k \alpha_{k,j}) w_j \in M_{k-1}.$$

This concludes the induction. $\square$

Definition 1.2.2 – (Full) Lattice

A **(full) lattice** in $\mathbb{R}^n$ is a discrete sugroup of $\mathbb{R}^n$ which has rank n.

Remark: For our intents and purposes, a **lattice** is always **full**.

Remark (Bases of Lattices)**:** If Λ is a lattice in $\mathbb{R}^n$, then by the previous lemma, there exists a basis $\{v_1, \dots, v_n\}$ such that $\Lambda = \{c_1 v_1 + \cdots + c_n v_n : c_1, \dots, c_n \in \mathbb{Z}\}$.

Definition 1.2.3 – Determinant of a Lattice

We define the **determinant** of Λ to be

$$\det(\Lambda) = |\det(v_1 \cdots v_n)|\,.$$

Recall that $\mathrm{GL}_n(\mathbb{Z})$ is the multiplicative group of $n \times n$-matrices over $\mathbb{Z}$ with determinant ± 1.

Lemma 1.2.4 (Lattice Basis Transformation)

Let $\Lambda \subseteq \mathbb{R}^n$ be a lattice. Suppose that $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_n\}$ are two bases of Λ. Then there exists $u = (u_{i,j})_{i,j \in \{1,\dots,n\}} \in \mathrm{GL}_n(\mathbb{Z})$ such that

$$w_i = \sum_{j=1}^{n} u_{i,j} v_j \ \forall_{1 \le i \le n}.$$

In particular,

$$|\det(w_1 \cdots w_n)| = |\det(v_1 \cdots v_n)|.$$

Proof. Let us define $U = (u_{i,j})_{i,j \in \{1,\dots,n\}}$ by

$$w_i = \sum_{j=1}^{n} u_{i,j} v_j \ \forall_{1 \le i \le n}.$$

Since $w_i \in \Lambda$ and $\{v_1, \dots, v_n\}$ is a basis of Λ, we get $u_{i,j} \in \mathbb{Z}\ \forall_{1 \le i,j \le n}$, which ensures $\det(U) \in \mathbb{Z}$. By the same argument, we define $\widetilde{U} = (\widetilde{u}_{i,j})_{i,j \in \{1,\dots,n\}}$ analogously. Note that $U^{-1} = \widetilde{U}$. Hence, $\det(\widetilde{U}) = \det(U) = \pm 1$. □

Let M, L be two lattices in $\mathbb{R}^n$ with $M \subseteq L$. Further let $\{v_1, \dots, v_n\}$ be a basis of L and $\{w_1, \dots, w_m\}$ be a basis of M. Define U_0 as the $n \times n$ matrix that fulfils

$$(w_1 \cdots w_n) = (v_1 \cdots v_n) U_0.$$

Then, we have $U_0 \in M_n(\mathbb{Z})$. We define the **index** of M in L by

$$[L : M] = |\det U_0| = \left| \frac{\det M}{\det L} \right|.$$

Note: This does not depend on the choice of bases of M and L.

Example. Let

$$\Lambda_0 = \mathrm{Span}_{\mathbb{Z}}\left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\} \text{ and } \Lambda = \mathrm{Span}_{\mathbb{Z}}\left\{\begin{pmatrix}2\\0\end{pmatrix}, \begin{pmatrix}0\\2\end{pmatrix}\right\}.$$ □

Lecture 2 (17th April 2026)

Lecture 3 (22nd April 2026)

1.3 Convex Bodies

Definition 1.3.1 – Convex Subset of $\mathbb{R}^n$

A set $C \subseteq \mathbb{R}^n$ is **convex** if given any $x, y \in C$, we have

$$\{tx + (1-t)y : t \in [0,1]\} \subseteq C.$$

Definition 1.3.2 – Central Symmetric Convex Body in $\mathbb{R}^n$

A **central symmetric convex body** in $\mathbb{R}^n$ is a closed bounded convex $C \subseteq \mathbb{R}^n$ fulfilling $0 \in C$ and $x \in C \iff -x \in C$.

Example. Consider any norm on $\mathbb{R}^n$. Then the set

$$B_1(0) = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$$

is a central symmetric convex body. □

Let $C \subseteq \mathbb{R}^n$ be a central symmetric convex body. Define

$$\|x\|_C = \min\{\lambda \in \mathbb{R}_{\geq 0} : x \in \lambda C\},$$

where $\lambda C = \{\lambda x : x \in C\}$.

Lemma 1.3.3 (Norm properties of $\|\cdot\|_C$)

The map $\|\cdot\|_C$ is

(i) well-defined,

(ii) a norm on $\mathbb{R}^n$ and

(iii) $\lambda C = \{x \in \mathbb{R}^n : \|x\|_C \leq \lambda\}$.

Proof. We'll prove (i). Proving the other two statements is left to the reader as an exercise. Clearly, $\|0\|_C = 0$. Let $x \in \mathbb{R}^n \setminus \{0\}$. Consider the set

$$S = \{\lambda \in \mathbb{R}_{\geq 0} : x \in \lambda C\}.$$

Since 0 is in the interior of C, there exists $\delta > 0$ such that

$$\{z \in \mathbb{R}^n : \|z\|_2 \leq \delta\} \subseteq C.$$

Hence, we have

$$x \in \frac{\|x\|_2}{\delta} C, \text{ which implies } S \neq \varnothing.$$

Let $0 \leq \mu = \inf_{\lambda \in S} \lambda$. By the definition of the infimum, we have

$$x \in \left(\mu + \frac{1}{m}\right) C \ \forall_{m \in \mathbb{N}_{\geq 1}}.$$

Thus, we have

$$\left(\mu + \frac{1}{m}\right)^{-1} x \in C.$$

Since C is bounded, it must be that $\mu > 0$. Furthermore, since C is closed and

$$\left(\mu + \frac{1}{m}\right)^{-1} x \in C \ \forall_{m \in \mathbb{N}_{\geq 1}},$$

we have

$$\lim_{m \to \infty} \left(\mu + \frac{1}{m}\right)^{-1} x = \frac{1}{\mu} x \in C,$$

which implies $x \in \mu C$, i.e. $\mu \in S$. □

1.4 Minkowski's First Theorem

Given a measurable $S \subseteq \mathbb{R}^n$, we define the n-dimensional volume $\mathrm{vol}(S) \in \overline{\mathbb{R}_{\geq 0}}$ as $\lambda^n(S)$, the n-dimensional Lebesgue measure of S.
Some key properties:

- All open and closed sets are measurable.
- Bounded measurable sets have finite volume.
- For measurable S,

  $$\mathrm{vol}(S) = \int_S 1 \, \mathrm{d}x_1 \cdots \mathrm{d}x_n,$$

  if this integral (the n-fold Riemann integral) is defined.
- The set of all measurable sets in $\mathbb{R}^n$ forms a σ-algebra over $\mathbb{R}$.
- If $\varphi : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation and $S \subseteq \mathbb{R}^n$ is measurable, then $\varphi(S)$ is measurable with

  $$\mathrm{vol}(\varphi(S)) = |\det \varphi| \mathrm{vol}(S).$$
- The function vol is σ-additive.

Let Λ be a full lattice inside of $\mathbb{R}^n$ and $\{v_1, \ldots, v_n\}$ be a basis. Then

$$F = \{\alpha_1 v_1 + \cdots + \alpha_n v_n : 0 \leq \alpha_1, \ldots, \alpha_n < 1\}$$

is a fundamental parallel piped of Λ. Then

$$\mathbb{R}^n = \bigcup_{u \in \Lambda} u + F,$$

where the $u + F$ are pairwise disjoint. Also,

$$\mathrm{vol}(F) = \int_{[0,1)^n} |\det[v_1 \cdots v_n]| \, \mathrm{d}\alpha_1 \cdots \mathrm{d}\alpha_n = \det \Lambda.$$

Theorem 1.4.1 (Minkowski's First Convex Body Theorem, 1896)

Let C be a central symmetric convex body in $\mathbb{R}^n$ and Λ a lattice. Suppose that

$$\mathrm{vol}(C) \geq 2^n \det(\Lambda).$$

Then $C \cap (\Lambda \setminus \{0\}) \neq \varnothing$.

Proof. First assume that $\mathrm{vol}(C) > 2^n \det(\Lambda)$. Then the set $1/2 C$ satisfies

$$\mathrm{vol}\left(\frac{1}{2}C\right) = \frac{1}{2^n}\mathrm{vol}(C) > \det \Lambda.$$

Let F be a fundamental parallel piped of Λ and

$$S_a = \frac{1}{2}C \cap (a + F) \ \forall_{a \in \Lambda}.$$

Then,

$$\sum_{a \in \Lambda} \mathrm{vol}(S_a) = \mathrm{vol}\left(\frac{1}{2}C\right) > \det \Lambda.$$

Let us define

$$\begin{aligned} T_a = & \quad -a + S_a \\ = & \quad \left(-a + \frac{1}{2}C\right) \cap F \subseteq F. \end{aligned}$$

Now,

$$\sum_{a \in \Lambda} \mathrm{vol}(T_a) = \sum_{a \in \Lambda} \mathrm{vol}(S_a) > \det \Lambda = \mathrm{vol}(F).$$

We proved the existence of a collection of subsets of F whose sum of volumes is greater than $\mathrm{vol}(F)$. Therefore, there exists $a \neq b \in \Lambda$ such that $T_a \cap T_b \neq \varnothing$. Let $u \in T_a \cap T_b$. There exists $x, y \in 1/2 C$ such that $x - a = u = y - b$.
This implies that $x - y = a - b \in \Lambda \setminus \{0\}$. We also know that $2x, 2y \in C$. Hence, we have $-2y \in C$, which implies

$$\frac{1}{2}(2x) + \frac{1}{2}(-2y) = x - y \in C.$$

Next, let us suppose $\mathrm{vol}(C) = 2^n \det(\Lambda)$. Let $m \in \mathbb{N}_{\geq 1}$. Since

$$\mathrm{vol}\left(\left(1 + \frac{1}{m}\right) C\right) > 2^n \det(\Lambda),$$

it follows from the previous case that there exists

$$0 \neq x_m \in (1 + 1/m)C \cap \Lambda \subseteq 2C \cap \Lambda \ \forall_{m \in \mathbb{N}_{\geq 1}}.$$

We know that $2C$ is bounded, so $2C \cap \Lambda$ is finite. Therefore, there exists an infinite sequence $m_1 < \cdots$ such that $0 \neq x_0 = x_{m_1} = x_{m_2} = \ldots$. Since $x_0 = x_{m_i} \in ((1 + 1/m_i)C) \cap \Lambda$, we have

$$\left(1 + \frac{1}{m}\right)^{-1} x_0 \in C \ \forall_{i \in \mathbb{N}_{\geq 1}}.$$

Hence, $0 \neq x_0 \in \Lambda \cap C$. □

Lemma 1.4.2 (Cardinality Formula for Lattice-Body Intersections)

Let C be a central symmetric convex body and Λ a full lattice in $\mathbb{R}^n$. Then

$$\#(xC \cap \Lambda) = \frac{\mathrm{vol}(C)}{\det(\Lambda)} x^n + \mathcal{O}(x^{n-1}) \ \forall_{x \geq 1}.$$

Proof. Let $N(x) = \#(xC \cap \Lambda)$. Consider

$$S = \bigcup_{u \in xC \cap \Lambda} u + F.$$

Since this is a disjoint union, we have

$$\mathrm{vol}(S) = N(x)\mathrm{vol}(F).$$

As F is bounded,

$$\exists_{\sigma > 0} : \|y\|_C \leq \sigma \ \forall_{y \in F}.$$

Let $x > \sigma$.

Claim 1.4.3

We have

$$(x - \sigma)C \subseteq S \subseteq (x + \sigma)C.$$

The result follows from this claim by taking the volume

$$(x - \sigma)^n \mathrm{vol}(C) \leq \mathrm{vol}(S) \leq (x + \sigma)^n \mathrm{vol}(C),$$

which can be rewritten as

$$\mathrm{vol}(S) = x^n \mathrm{vol}(C) + \mathcal{O}(x^{n-1}).$$

Then just divide by $\mathrm{vol}(F)$.

Proof of the claim. Given any $x \in \mathbb{R}^n$, let us write $x = u + y$ with $u \in \Lambda$ and $y \in F$. Then $(x - \sigma)C \subseteq S$, because $\|x\|_C \leq x - \sigma$, which implies $\|u\| \leq x$ (use the triangle inequality $\|u\|_C \leq \|x - u\| + \|x\|_C = \|y\|_C + \|x\|_C \leq \sigma + x - \sigma = x$). Similarly, $S \subseteq (x + \sigma)C$ follows from $\|u\|_C \leq x$, which implies $\|x\|_C \leq x + \sigma$ (again triangle inequality). □

This completes the proof. □

Lecture 3 (22nd April 2026)

. . . . . .

Lecture 4 (24th April 2026)

We will study another proof of Minkowski's first theorem 1.4.1 using Lemma 1.4.2.

Second proof of Theorem 1.4.1. Let $m \in \mathbb{N}_{\geq 1}$. We divide Λ into congruence classes modulo $2m$. Each $x \in \Lambda$ is written as

$$x = x_1 v_1 + \cdots + x_n v_n,$$

where $\{v_1, \ldots, v_n\}$ is a basis of Λ. Consider

$$\Lambda_a = \{x_1 v_1 + \cdots + x_n v_n : x_i \equiv a_i \mod 2m (1 \leq i \leq n)\},$$

for each $a \in \{0, \ldots, 2m - 1\}^n$. Using Lemma 1.4.2, we have

$$\begin{aligned} \#(mC \cap \Lambda) = & \frac{\mathrm{vol}(C)}{\det(\Lambda)} m^n + \mathcal{O}(m^{n-1}) \\ > & 2^n (1 + \varepsilon) m^n, \end{aligned}$$

provided that m is sufficiently large. Hence, there exists $a \in \{0, \ldots, 2m - 1\}^n$ such that $\Lambda_a \cap mC$ contains at least two distinct vectors u, v. Then $0 \neq 2m(u - v) \in \Lambda$. Since we have $u, v \in mC$, we also have $u, -v \in mC$, which implies $1/2u - 1/2v \in mC$. Finally, $1/2m(u - v) \in C$, which completes the proof. □

Corollary 1.4.4

Let $\alpha_{i,j} \in \mathbb{R}$ for $1 \leq i, j \leq n$ and $L_i(x) = \alpha_{i,1} x_1 + \cdots + \alpha_{i,n} x_n$. Suppose that $0 < |\det[a_{i,j}]| \leq A_1 \cdots A_n$, where $A_1, \ldots, A_n > 0$.
Then, there exists $x \in \mathbb{Z}^n \setminus \{0\}$ such that

$$|L_i(x)| \leq A_i \ \forall_{i \in \{1, \ldots, n\}}.$$

Proof. Let $\mathfrak{B} = [-1, 1]^n$ and $\Lambda = \left\{ \left(\frac{L_1(x)}{A_1}, \ldots, \frac{L_n(x)}{A_n} \right) : x \in \mathbb{Z}^n \right\}$. Then this is a lattice inside $\mathbb{R}^n$. Also,

$$\left\{ \begin{pmatrix} \frac{\alpha_{1,1}}{A_1} & \cdots & \frac{\alpha_{1,n}}{A_1} \\ \vdots & & \vdots \\ \frac{\alpha_{n,1}}{A_n} & \cdots & \frac{\alpha_{n,n}}{A_n} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : x \in \mathbb{Z}^n \right\}.$$

We have

$$\det \Lambda = \frac{|\det[a_{i,j}]|}{A_1 \cdots A_n} \leq 1,$$

which implies

$$\frac{\mathrm{vol}(B)}{\det(\Lambda)} = \frac{2^n}{\det \Lambda} \geq 2^n.$$

By Minkowski's first theorem, there exists $0 \neq x \in \Lambda \cap \mathfrak{B}$. □

Corollary 1.4.5

Let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, not all in $\mathbb{Q}$. Then,

$$\#\{(x_1, \ldots, x_n, y) \in \mathbb{Z}^{n+1}_{\mathrm{prim}} : y > 0, \left|\alpha_i + \frac{x_i}{y}\right| \leq y^{-1-1/n} \ \forall_{1 \leq i \leq n}.$$

(Denote $\mathbb{Z}^n_{\mathrm{prim}} = \{(x_1, \ldots, x_n) \in \mathbb{Z}^n : \gcd(x_1, \ldots, x_n) = 1\}$.)

Proof. Let $Q > 1$. Consider the set

$$\{(x_1, \ldots, x_n, y) \in \mathbb{Z}^{n+1} : |x_i - \alpha_i y| \leq Q^{-1/n}, 0 < |y| \leq Q\}.$$

Let

$$\begin{pmatrix} L_1(x,y) \\ \vdots \\ L_n(x,y) \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 & -\alpha_1 \\ 0 & \ddots & \ddots & \vdots & \vdots \\ \vdots & \ddots & \ddots & 0 & -\alpha_{n-2} \\ \vdots & \cdots & \ddots & \ddots & -\alpha_{n-1} \\ 0 & \cdots & \cdots & 0 & 1 \end{pmatrix}}_{:=\widehat{A}} \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y \end{pmatrix}.$$

We use Corollary 1.4.4 (with $A_1 = \cdots = A_n = Q^{-1/n}$ and $A_{n+1} = Q$). Then there exists $(x, y) \in \mathbb{Z}^{n+1} \setminus \{0\}$ such that

$$\begin{aligned} |x_i - \alpha_i y| &\leq Q^{-1/n} \\ |y| &\leq \quad Q \ \forall_{1 \leq i \leq n}. \end{aligned}$$

(since $A_1 \cdots A_{n+1} = 1 = \det \widehat{A}$). We know that $y \neq 0$ because $y = 0$ would imply $(x, y) = 0$. If $\gcd(x_1, \ldots, x_n, y) > 1$, then divide through by the gcd. Also, if $y < 0$, divide throuhg by (-1). Now, for any $Q > 1$, there exists $(x_1, \ldots, x_n, y) \in \mathbb{Z}^{n+1}_{\mathrm{prim}} \setminus \{0\}$ such that $0 < y \leq Q$ and $|x_i - \alpha_i y| \leq Q^{-1/n} \ \forall_{1 \leq i \leq n}$. The assertion now follows through reasoning analogously to Corollary 1.1.3. □

Given $\theta \in \mathbb{R}$, we denote

$$\|\theta\| = \min\{|\theta - n| : n \in \mathbb{Z}\}.$$

From Corollary 1.4.5, for $\alpha_1, \alpha_2 \in \mathbb{R}$, not both in $\mathbb{Q}$, we get

$$\begin{aligned} &\#\{y \in \mathbb{N}_{\geq 1} : \|\alpha_1 y\| \leq y^{-1/2}, \|\alpha_2 y\| \leq y^{-1/2}\} = \aleph_0 \\ \implies &\#\{y \in \mathbb{N}_{\geq 1} : y\|\alpha_1 y\|\|\alpha_2 y| \leq 1\} = \aleph_0 \end{aligned}$$

(note that this also holds for $\alpha_1, \alpha_2 \in \mathbb{Q}$).

Hypothesis 1.4.6 (Littlewood)

Let $\alpha_1, \alpha_2 \in \mathbb{R}$. For any $\varepsilon > 0$, there exists $y \in \mathbb{N}_{\geq 1}$ such that

$$y\|\alpha_1\|\|\alpha_2\| < \varepsilon.$$

1.5 Minkowski's Second Theorem

Let Λ be a lattice and C a central symmetric convex body in $\mathbb{R}^n$. The successive minima $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ of Λ with respect to C are defined as

$$\lambda_i = \min\{\lambda \in \mathbb{R}_{\geq 0} : \dim \operatorname{Span}_{\mathbb{R}}(\lambda C \cap \Lambda) \geq i\} \ \forall_{1 \leq i \leq n}.$$

Lemma 1.5.1

The successive $0 < \lambda_1 \leq \cdots \leq \lambda_n < \infty$ are well-defined. Also, there are $v_1, \ldots, v_n \in \Lambda$, which are linearly independent over $\mathbb{R}$, such that $v_i \in \lambda_i C \ \forall_{1 \leq i \leq n}$.

Proof. Recall that

$$\|x\|_C = \min\{\lambda \in \mathbb{R}_{\geq 0} Lx \in \lambda C\} \text{ and } \lambda C = \{x \in \mathbb{R}^n : \|x\|_C \leq \lambda\}.$$

Let $\Lambda \setminus \{0\} = \{x_1, \ldots\}$, where $0 < \|x_1\|_C \leq \ldots$. Also note that given any $m \in \mathbb{N}_{\geq 1}$, there are finitely many i such that $m - 1 \leq \|x_i\|_C \leq m$ (because mC is bounded and λ is discrete). Let $\lambda_1 = \|x_1\|_C$. For $i \in \{2, \ldots, n\}$, define

$$k_i = \min\{k \in \mathbb{N}_{\geq 1} : \dim \operatorname{Span}_{\mathbb{R}}\{x_1, \ldots, x_k\} = i\} \text{ and set } v_i = x_k \text{ as well as } \lambda_i = \|x_i\|_C.$$

Clearly, $v_i \in \lambda_i C \cap \Lambda$ and $0 < \lambda_1 \leq \cdots \leq \lambda_n < 0$. Finally, we show that these are indeed the successive minima: By construction, the

$$v_1, \ldots, v_i \in \lambda_i C \cap \Lambda \text{ are linearly independent over } \mathbb{R}.$$

Take $0 < \lambda < \lambda_i$ and let ℓ be the largest number such that $x_\ell \in \lambda C \cap \Lambda$. Then, we have $\ell < k_i$, and by the definition of the k_i, it follows that

$$\dim \operatorname{Span}_{\mathbb{R}}(\lambda C \cap \Lambda) = \dim \operatorname{Span}_{\mathbb{R}}\{x_1, \ldots, x_\ell\} < i.$$

This shows that λ_i is indeed the successive minimum. □

Theorem 1.5.2 (Minkowski's Second Convex Body Theorem)

Let *Lambda* $\subseteq \mathbb{R}^n$ be a lattice and $C \subseteq \mathbb{R}^n$ a central symmetric convex body. Further assume that $\lambda_i, \ldots, \lambda_n$ are the successive minima of Λ with respect to C. Then, we have

$$\frac{2^n}{n!} \frac{\det \Lambda}{\operatorname{vol}(C)} \leq \lambda_1 \cdots \lambda_n \leq 2^n \frac{\det \Lambda}{\operatorname{vol}(C)}.$$

Before we prove this, let us study a couple of examples to highlight the importance of this result.

Example. Let $n \geq 2$ and

$$\mathfrak{B} = B_1^n(0).$$

Further let $\Lambda \subseteq \mathbb{R}^n$ be a lattice and $\lambda_1, \ldots, \lambda_n$ be the successive minima of Λ with respect to $\mathfrak{B}$. It follows from Lemma 1.5.1 that there exist $x_1, \ldots, x_n \in \Lambda$, which are linearly independent over $\mathbb{R}$ and fulfil $x_i \in \lambda_i \mathfrak{B} \cap \Lambda$. Hence, we have $\|x_i\|_{\mathfrak{B}} = \lambda_i \iff |x_i| = \lambda_i$. Theorem 1.5.2

$$\prod_{i=1}^{n} |x_i| \leq \frac{2^n \det \Lambda}{V_n},$$

where

$$V_n = \operatorname{vol}(\mathfrak{B}) = \begin{cases} 1 & \text{if } n = 1, \\ \pi & \text{if } n = 2, \\ \frac{2\pi}{n} V_{n-2} & \text{if } n \geq 3. \end{cases}$$

Note that $\{x_1, \ldots, x_n\}$ might not be a basis of Λ. □

Example. Let $\Lambda = \mathbb{Z}^n$. Take any real $0 < \lambda_1 < \cdots < \lambda_n$. Define $C = \{x \in \mathbb{R}^n : |x_i| \leq {}^1\!/_{\lambda_i} \ \forall_{1 \leq i \leq n}\}$. Then C is a central symmetric convex body with

$$\operatorname{vol}(C) = \frac{2^n}{\lambda_1 \cdots \lambda_n}. \tag{$*$}$$

Let us now show that the λ_j are the successive minima of Λ with respect to C. For $\lambda \geq 0$, we have

$$\lambda C = \{x \in \mathbb{R}^n : |x_i| \leq {}^{\lambda}\!/_{\lambda_i} \ \forall_{1 \leq i \leq n}\}.$$

Then $e_1, \ldots, e_j \in \lambda_j C$ are linearly independent over $\mathbb{R}$. Suppose that $0 < \lambda < \lambda_j$ and $y \in \mathbb{Z}^n \cap \lambda C$. Necessarily, $y_{i+1} = \cdots = y_n = 0$. Therefore,

$$\mathbb{Z}^n \cap \lambda C \subseteq \mathbb{Z}^{j-i} \times \{(0, \ldots, 0) \in \mathbb{Z}^{n-j+1}\}.$$

This can contain j linearly independent vectors. Hence, λ_j is the successive minimum. Thus,

$$\lambda_1 \cdots \lambda_n \overset{(*)}{=} \frac{2^n}{\operatorname{vol}(C)} = \frac{2^n \det \Lambda}{\operatorname{vol}(C)}.$$

This proves that the upper bound is sharp. □

Example. Take $\Lambda = \mathbb{Z}^n$,

$$C = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^{n} |x_i| \leq 1 \right\}.$$

Exercise 1.1

Show that $\operatorname{vol}(C = \frac{2^n}{n!})$ and $\lambda_1 = \cdots = \lambda_n = 1$.

This proves that the lower bound in Theorem 1.5.2 is sharp. □

Lecture 4 (24th April 2026)

. . . . . .

Lecture 5 (29th April 2026)

Appendix

List of Definitions

1.1.1 $\mathbb{Z}^2_{\text{prim}}$ 2
1.2.2 (Full) Lattice 6
1.2.3 Determinant of a Lattice 6
1.3.1 Convex Subset of $\mathbb{R}^n$ 7
1.3.2 Central Symmetric Convex Body in $\mathbb{R}^n$ 7

List of Theorems

1.1.2 Dirichlet, 1842 2
1.1.5 Roth, 1955 3
1.1.6 Thue, 1909 3
1.4.1 Minkowski's First Convex Body Theorem, 1896 9
1.5.2 Minkowski's Second Convex Body Theorem 13

List of Lemmata

1.2.1 Lattice Basis Construction 4
1.2.4 Lattice Basis Transformation 6
1.3.3 Norm properties of $\|\cdot\|_C$ 7
1.4.2 Cardinality Formula for Lattice-Body Intersections 10
1.5.1 13

Index

$N(f)$, 1
$N^*(f)$, 1
$\mathbb{Z}^2_{\text{prim}}$, 2
(Full) Lattice, 6

Basis of a Discrete Subgroup of $\mathbb{R}^n$, 4

Central Symmetric Convex Body in $\mathbb{R}^n$, 7

Convex Subset of $\mathbb{R}^n$, 7

Determinant of a Lattice, 6
Discrete Subgroup of $\mathbb{R}^n$, 4
Discrete Subset of $\mathbb{R}^n$, 4

Index of a Sublattice in a Lattice, 6

Rank of a Discrete Subgroup of $\mathbb{R}^n$, 4